Message broker
[Container: Mosquitto]

rooms.json
[Config file]

Occupancy simulator
[Software System]

simulated rooms

Decision maker [Container: Python, con-dec.py]

…/dec readings, …/actuator-status acks

Controller registry
[Component]
build_controller(): one temp + one CO_2 controller per simulated room

routes

clock [REST]

Topic router
[Component]
on_message(): topic → controller
(20 subscriptions for 5 rooms)

Health watchdog
[Component]
1 s loop over all controllers

staleness check

ack

reading

timeout check → resend

Room controller (×2 per room)

Sensor health monitor
[Component]
stale > 15 s or 5 identical readings → fail-safe

Setpoint scheduler
[Component]
21 °C 06–18, else 17 °C; CO_2 fixed 900 ppm

Actuator ack monitor
[Component]
ack within 10 s or retry

…/ac commands

healthy reading

setpoint

PID regulator
[Component]
clamped output, anti-windup

fail-safe output

wattage

Command publisher
[Component]
publish_wattage(): watt + command_id